# Lecture 13

## Metric, Normed, and Hilbert Spaces

Every space so far has been finite-dimensional. Quantum mechanics is not: a wavefunction lives in an infinite-dimensional space of functions, and to speak of convergence, limits, and complete orthonormal expansions there we need *analysis* layered on top of linear algebra. This lecture supplies it. We build metric and normed spaces, isolate the crucial property of *completeness* (Banach spaces), and arrive at *Hilbert spaces*—complete inner product spaces—with the two that run through all of physics, $\ell^2$ and $L^2$. Orthonormal bases, Bessel's inequality, Parseval's identity, Fourier series, and the Riesz theorem all survive the passage to infinite dimensions, provided we respect completeness and continuity. This opens Part III.

### A note on rigor in Part III

Almost every result so far has been proved in full; the handful of exceptions (triangularizability and the generalized-eigenspace decomposition in Lecture 5, Jordan form in Lecture 6, and the two perturbation results in Lecture 10) already carry the tags described here. Part III layers analysis onto linear algebra, and more of its results need more of it than this course assumes, so from here each theorem heading carries its status. An untagged result is proved here, as before; one marked sketched gives the central idea but omits a technical step (usually a convergence or completeness argument); and one marked cited is stated and motivated, with its full proof left to a functional-analysis course (Hall, in the syllabus references). For a cited result the goal is to know what it says and how to use it—not to reproduce the proof.

### Learning objectives

By the end of this lecture you will be able to:

- Distinguish metrics, norms, and inner-product norms (via the parallelogram law).
- Define Cauchy sequences and completeness; tell Banach from Hilbert spaces.
- Work in the model spaces $\ell^2$ and $L^2$ and compute Fourier expansions.
- Use a Hilbert basis with Bessel, Parseval, and completeness; state Riesz–Fréchet.

## 1 Metric and normed spaces

**Definition 13.1 (Metric space).** A *metric* on a set $X$ is a function $d\colon X \times X \to [0,\infty)$ with $d(x,y) = 0 \iff x = y$, $d(x,y) = d(y,x)$, and the triangle inequality $d(x,z) \le d(x,y) + d(y,z)$. The pair $(X,d)$ is a *metric space*.

**Definition 13.2 (Normed space).** A *norm* on a vector space $V$ assigns $\|v\| \ge 0$ with $\|v\| = 0 \iff v = 0$, $\|av\| = |a|\,\|v\|$, and $\|u+v\| \le \|u\| + \|v\|$. Every norm induces a metric $d(u,v) = \|u-v\|$.

A sequence $v_n$ *converges* to $v$ if $\|v_n - v\| \to 0$, and is *Cauchy* if $\|v_m - v_n\| \to 0$ as $m, n \to \infty$. Convergent sequences are always Cauchy; the converse is the content of completeness.

**Example 13.3 (Norms).** On $\mathbb{C}^n$ the Euclidean norm $\|x\|_2 = \sqrt{\sum |x_i|^2}$ comes from the inner product; the taxicab norm $\|x\|_1 = \sum |x_i|$ and the sup norm $\|x\|_\infty = \max_i |x_i|$ do not. On

$C[a,b]$ the uniform norm $\|f\|_\infty = \sup|f|$ and the $L^2$ norm $\|f\|_2 = \sqrt{\int |f|^2}$ are genuinely different. ◁

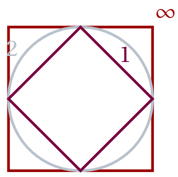


Figure 1: Unit balls in $\mathbb{R}^2$ for $\|\cdot\|_1$ (diamond), $\|\cdot\|_2$ (circle), and $\|\cdot\|_\infty$ (square). Only the round Euclidean ball comes from an inner product.

**Example 13.4 (Cauchy, but where is the limit?).** In $\mathbb{Q}$ the sequence $3,\ 3.1,\ 3.14,\ 3.141, \ldots$ is Cauchy, but its limit $\pi$ is irrational: $\mathbb{Q}$ is incomplete. Filling the gaps gives $\mathbb{R}$. The same gap-filling turns an inner product space into a Hilbert space. ◁

**Example 13.5 (Other metrics).** Not every metric comes from a norm. The *discrete metric* $d(x,y) = 1$ for $x \neq y$ records only "same or different"; the $p$-norms $\|x\|_p = (\sum |x_i|^p)^{1/p}$ interpolate between taxicab ($p = 1$) and Euclidean ($p = 2$), and only $p = 2$ comes from an inner product. ◁

**Example 13.6 (Checking the triangle inequality).** The triangle inequality is Cauchy–Schwarz in disguise: squaring, $\|u+v\|^2 = \|u\|^2 + 2\Re\langle u, v\rangle + \|v\|^2 \leq \|u\|^2 + 2\|u\|\|v\| + \|v\|^2 = (\|u\| + \|v\|)^2$. The same one line proves it in every inner product space, finite- or infinite-dimensional. ◁

## 2 Completeness: Banach and Hilbert spaces

**Definition 13.7 (Complete; Banach; Hilbert).** A metric space is *complete* if every Cauchy sequence converges (to a point of the space). A complete normed space is a *Banach space*. An inner product gives a norm $\|v\| = \sqrt{\langle v, v\rangle}$; a complete inner product space is a *Hilbert space* $\mathcal{H}$.

Completeness is the assurance that limits one expects to exist actually do—there are no "missing points." Every finite-dimensional normed space is complete, so in finite dimensions the distinction never arose. It becomes essential in infinite dimensions, where natural spaces can have holes.

**Example 13.8 (An incomplete inner product space).** The continuous functions $C[-1,1]$ with $\langle f, g\rangle = \int_{-1}^{1} \overline{f} g$ form an inner product space that is *not* complete: continuous functions approximating a step are Cauchy in the $L^2$ norm yet converge to a discontinuous limit outside $C[-1,1]$. Its completion is $L^2[-1,1]$. ◁

**Remark 13.9 (Which norms come from inner products).** A norm comes from an inner product if and only if it obeys the parallelogram law $\|u+v\|^2 + \|u-v\|^2 = 2\|u\|^2 + 2\|v\|^2$ (Lecture 7), in which case polarization recovers the inner product. (The forward direction is the computation of Lecture 7; the converse—that the parallelogram law forces the polarization formula to be an inner product—is the Jordan–von Neumann theorem, stated here without proof.) The sup norm fails it, so $C[a,b]$ with $\|\cdot\|_\infty$ is a Banach space but not a Hilbert space.

**Proposition 13.10 (Completeness via series · cited).** A normed space is complete if and only if every absolutely convergent series (one with $\sum_n \|v_n\| < \infty$) converges. This is the working test for completeness, and the reason infinite expansions behave well in a Hilbert space.

**Example 13.11 (Banach but not Hilbert).** The sequence space $\ell^1$ ($\sum |x_n| < \infty$, norm $\sum |x_n|$) is complete, hence Banach, but its norm violates the parallelogram law, so no inner product induces it. Among the $\ell^p$ spaces only $\ell^2$ is a Hilbert space. ◁

**Remark 13.12 (Finite dimensions are always complete).** Every finite-dimensional normed space is complete, and all norms on it are equivalent, so the analytic subtleties of this lecture are invisible there. They surface only in infinite dimensions—exactly where quantum mechanics lives.

**Remark 13.13 (The closed unit ball is no longer compact).** A second, sharper symptom of infinite dimension: in $\mathbb{R}^n$ every closed bounded set is compact (every sequence in it has a convergent subsequence), but in an infinite-dimensional Hilbert space the closed unit ball fails this. The proof is one line: an orthonormal sequence $e_1, e_2, \ldots$ lies in the unit ball, yet by the Pythagorean theorem

$$\|e_m - e_n\|^2 = \|e_m\|^2 + \|e_n\|^2 = 2 \qquad (m \neq n),$$

so all its terms stay a distance $\sqrt{2}$ apart and *no* subsequence can converge. "Closed and bounded" therefore no longer means compact, and the operators that retain finite-dimensional behavior will be exactly those that tame this failure—the *compact* operators of Lecture 14.

**Example 13.14 (Completing a space).** The polynomials on $[0,1]$ under the $L^2$ norm are incomplete, but they are dense (Weierstrass), so their completion is all of $L^2[0,1]$. Completion adds the limit of every Cauchy sequence—here, every square-integrable function. ◁

**Example 13.15 (The sup norm fails the parallelogram law).** Take $f = 1$ and $g(x) = x$ on $[0,1]$ in the sup norm: $\|f\|_\infty = \|g\|_\infty = 1$, $\|f+g\|_\infty = 2$, $\|f-g\|_\infty = 1$, so

$$\|f+g\|_\infty^2 + \|f-g\|_\infty^2 = 5 \neq 4 = 2\|f\|_\infty^2 + 2\|g\|_\infty^2.$$

The parallelogram law fails, confirming that the sup norm comes from no inner product. ◁

## 3 The Hilbert spaces of physics: $\ell^2$ and $L^2$

**Definition 13.16 ($\ell^2$ and $L^2$).** The sequence space $\ell^2$ consists of $(x_1, x_2, \ldots)$ with $\sum_n |x_n|^2 < \infty$, inner product $\langle x, y\rangle = \sum_n \overline{x_n} y_n$. The function space $L^2[a,b]$ consists of (equivalence classes of) functions with $\int_a^b |f|^2 < \infty$, inner product $\langle f, g\rangle = \int_a^b \overline{f}\, g$.

Both are Hilbert spaces—the completeness of $L^2$ is the Riesz–Fischer theorem—and both are *separable* (they admit a countable orthonormal basis). They are the state spaces of quantum mechanics: $\ell^2$ for discrete systems (energy levels, spin chains), $L^2$ for a particle on a line (the position wavefunction $\psi(x)$).

**Example 13.17 (Inside and outside $\ell^2$).** The sequence $(1, \frac{1}{2}, \frac{1}{3}, \ldots)$ lies in $\ell^2$ since $\sum 1/n^2 < \infty$, while the constant sequence $(1,1,1,\ldots)$ does not. The standard vectors $e_n$ (a 1 in slot $n$, zeros elsewhere) form a complete orthonormal basis, with $x = \sum_n x_n e_n$. ◁

**Example 13.18 (Wavefunctions).** A quantum state on the line is a $\psi \in L^2(\mathbb{R})$ with $\int|\psi|^2 = 1$, and $|\psi(x)|^2$ is the position probability density. The Gaussian $\psi(x) = \pi^{-1/4}e^{-x^2/2}$, the oscillator ground state, lies in $L^2(\mathbb{R})$; plane waves $e^{ikx}$ do not—a hint of the continuous spectrum to come. ◁

**Remark 13.19 (Separability).** $\ell^2$ and $L^2$ are *separable*: each has a countable complete orthonormal basis, so every state is a countable superposition. All separable infinite-dimensional Hilbert spaces are isomorphic, so $\ell^2$ and $L^2$ are the same Hilbert space in different clothes—a unitary map (expansion in a countable orthonormal basis; for $L^2[-\pi,\pi]$ this is the Fourier-coefficient map) carries one to the other.

**Example 13.20 (Normalizing a state).** The function $\psi(x) = e^{-|x|}$ on $\mathbb{R}$ has $\int_{-\infty}^{\infty} e^{-2|x|}\,dx = 1$, so it is already normalized in $L^2(\mathbb{R})$. Normalizability—membership in $L^2$—is the mathematical content of “the particle is somewhere.” ◁

**Example 13.21 (An $L^2$ inner product).** On $L^2[0,1]$, $\langle 1, x\rangle = \int_0^1 \overline{1}\cdot x\,dx = \frac{1}{2}$, while $\|1\|^2 = 1$ and $\|x\|^2 = \int_0^1 x^2 = \frac{1}{3}$. The functions 1 and $x$ are not orthogonal—Gram–Schmidt will fix that. ◁

**Example 13.22 (Cauchy–Schwarz for integrals).** In $L^2$, Cauchy–Schwarz reads $|\int \overline{f}g| \le \|f\|_2\|g\|_2$, that is $\left(\int|fg|\right)^2 \le \int|f|^2\int|g|^2$. It guarantees that the inner product $\int\overline{f}g$ converges whenever $f, g \in L^2$—the reason $L^2$ is closed under forming inner products. ◁

**Example 13.23 (Computing in $\ell^2$).** For $x = (1, \frac{1}{2}, \frac{1}{4}, \dots)$ and $y = (1, \frac{1}{3}, \frac{1}{9}, \dots)$, both in $\ell^2$,

$$\langle x, y\rangle = \sum_{n\ge 0}\left(\tfrac{1}{2}\right)^n\left(\tfrac{1}{3}\right)^n = \sum_{n\ge 0}\left(\tfrac{1}{6}\right)^n = \frac{1}{1-\frac{1}{6}} = \frac{6}{5}.$$

Geometric tails make $\ell^2$ computations concrete. ◁

**Example 13.24 (Normalizing on an interval).** To normalize $\psi(x) = x$ on $[0,1]$, compute $\|x\|^2 = \int_0^1 x^2\,dx = \frac{1}{3}$, so the normalized state is $\sqrt{3}\,x$. A constant on $[0,1]$ normalizes to 1, the uniform state. ◁

**Example 13.25 (Normalizing the Gaussian).** The Gaussian $\phi(x) = e^{-x^2/2}$ has

$$\|\phi\|^2 = \int_{-\infty}^{\infty} e^{-x^2}\,dx = \sqrt{\pi},$$

so the normalized oscillator ground state is $\pi^{-1/4}e^{-x^2/2}$, matching Example 13.18. Square-integrability is what keeps the normalization finite. ◁

**Example 13.26 (Number states).** The harmonic oscillator’s energy eigenstates $|n\rangle$ ($n \ge 0$) are a complete orthonormal basis, so its Hilbert space is a copy of $\ell^2$. A state $\sum_n c_n|n\rangle$ corresponds to the sequence $(c_n)$, which is square-summable; the ladder operator $a|n\rangle = \sqrt{n}\,|n-1\rangle$ is then an operator on $\ell^2$, the subject of Lecture 14. ◁

**Remark 13.27 (Any index set).** One can form $\ell^2(\Omega)$ over any set $\Omega$: families $(x_\omega)$ with $\sum|x_\omega|^2 < \infty$ (so at most countably many are nonzero). For countable $\Omega$ this is the separable $\ell^2$; an uncountable $\Omega$ gives a non-separable Hilbert space, rare in physics but logically available.

## 4 Orthonormal bases in infinite dimensions

An orthonormal sequence $e_1, e_2, \dots$ is a *complete orthonormal basis* (a *Hilbert basis*) if its closed span is all of $\mathcal{H}$. Completeness of the basis must not be confused with completeness of the space.

**Remark 13.28 ("Basis" needs rethinking).** The basis of Lecture 2—every vector a *finite* linear combination—is called a *Hamel basis*, and in an infinite-dimensional Hilbert space it becomes useless: one can show a Hamel basis of $\ell^2$ or $L^2$ is necessarily uncountable, and no explicit one can be written down. The right notion replaces finite sums by convergent infinite ones: a Hilbert basis is a *countable* orthonormal set whose finite combinations come arbitrarily close to every vector (closed span = $\mathcal{H}$). The standard vectors $e_n$ are a Hilbert basis of $\ell^2$ but not a Hamel basis—the sequence $(1, \frac{1}{2}, \frac{1}{4}, \dots)$ is a limit of finite combinations, not a finite combination itself. From here on, "basis" in an infinite-dimensional Hilbert space always means Hilbert basis.

**Theorem 13.29 (Expansion, Bessel, Parseval · sketched).** Let $\{e_n\}$ be orthonormal in a Hilbert space $\mathcal{H}$. For every $v$, Bessel's inequality holds,

$$\sum_n |\langle e_n, v\rangle|^2 \le \|v\|^2,$$

and the Fourier series $\sum_n \langle e_n, v\rangle\, e_n$ converges. The set is a complete orthonormal basis if and only if equality holds for all $v$ (Parseval), equivalently $v = \sum_n \langle e_n, v\rangle\, e_n$ for all $v$.

**Theorem 13.30 (Characterizing a complete basis · cited).** For an orthonormal set $\{e_n\}$ in a Hilbert space the following are equivalent: it is a complete orthonormal basis; it is maximal (no unit vector is orthogonal to every $e_n$); its closed span is the whole space; Parseval's identity holds for every vector.

The coefficients $\langle e_n, v\rangle$ are the Fourier coefficients, and the expansion is the abstract Fourier series of Lecture 8 carried to infinitely many terms; convergence is in the $L^2$ norm, not necessarily pointwise. The classical example is the trigonometric basis of $L^2[-\pi, \pi]$, whose expansion is the ordinary Fourier series.

**Example 13.31 (A Fourier series).** The functions $\frac{1}{\sqrt{2\pi}} e^{inx}$ ($n \in \mathbb{Z}$) form a complete orthonormal basis of $L^2[-\pi, \pi]$. Expanding $f(x) = x$ gives $x = 2\sum_{n\ge1} \frac{(-1)^{n+1}}{n} \sin nx$ (convergence in $L^2$), and Parseval's identity applied to it reproduces $\sum_{n\ge1} \frac{1}{n^2} = \frac{\pi^2}{6}$. ◁

**Theorem 13.32 (Projection onto a closed subspace · cited).** If $M$ is a closed subspace of a Hilbert space $\mathcal{H}$, every $v$ has a unique nearest point $Pv \in M$, with $v - Pv \perp M$; thus $\mathcal{H} = M \oplus M^\perp$ and $P$ is the orthogonal projection.

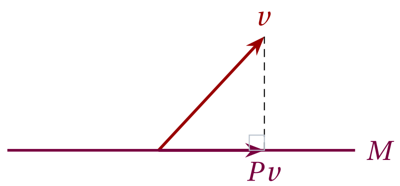


Figure 2: Projection onto a closed subspace: the nearest point $Pv$ is the foot of the perpendicular from $v$, and $v - Pv$ is orthogonal to all of $M$. Closedness (hence completeness) is what guarantees the nearest point exists.

**Example 13.33 (Truncation is the best approximation).** Among all combinations of $e_1, \dots, e_N$, the truncated Fourier sum $\sum_{n=1}^{N} \langle e_n, v\rangle\, e_n$ is the closest to $v$ (Lecture 8, now in infinite dimensions). The squared error $\|v\|^2 - \sum_{n=1}^{N} |\langle e_n, v\rangle|^2$ shrinks to zero exactly when the basis is complete. ◁

**Example 13.34 (A closed subspace and its complement).** In $L^2[-1,1]$ the even functions form a closed subspace $M$; its orthogonal complement $M^\perp$ is the odd functions, and every $f$ splits uniquely as even-plus-odd, $f = \frac{1}{2}(f(x)+f(-x)) + \frac{1}{2}(f(x)-f(-x))$. This is $\mathcal{H} = M \oplus M^\perp$ in action. ◁

**Example 13.35 (Orthogonal polynomials).** Gram–Schmidt applied to the monomials yields the Legendre polynomials on $L^2[-1,1]$ and the Hermite functions on $L^2(\mathbb{R})$ with a Gaussian weight, each a complete orthonormal basis. So a wavefunction expands in energy eigenstates—the spectral expansions of the next lecture. ◁

**Remark 13.36 (The Fourier transform).** The Fourier transform is a *unitary* map $L^2(\mathbb{R}) \to L^2(\mathbb{R})$ (Plancherel: it preserves the $L^2$ norm). It carries the position representation $\psi(x)$ to the momentum representation $\tilde{\psi}(p)$—two unitarily equivalent descriptions of one state, the infinite-dimensional analogue of a change of orthonormal basis.

**Example 13.37 (First Legendre polynomials).** On $L^2[-1,1]$, Gram–Schmidt on $1, x, x^2$ gives (up to normalization) $p_0 = 1$ and $p_1 = x$ (already orthogonal to 1 since $\int_{-1}^{1} x = 0$). For the third, $\int_{-1}^{1} x^2 = \frac{2}{3}$ and $\int_{-1}^{1} 1 = 2$ give the component along 1 as $\frac{1}{3}$, so $p_2 = x^2 - \frac{1}{3}$. These are the Legendre polynomials. ◁

**Remark 13.38 (Modes of convergence).** Convergence in $L^2$ is weaker than pointwise: a Fourier series converges to $f$ in the $L^2$ norm for every $f \in L^2$, yet may fail at individual points. For physics the $L^2$ sense is the relevant one—it is the convergence of probability amplitudes.

**Example 13.39 (Parseval as an energy identity).** For the Fourier basis, Parseval reads $\int |f|^2 = \sum_n |c_n|^2$ with $c_n$ the Fourier coefficients: a signal's total energy is the sum of the energies of its frequency components. Applied to $f(x) = x$ on $[-\pi, \pi]$ it again gives $\sum 1/n^2 = \pi^2/6$. ◁

**Example 13.40 (A two-term approximation).** The best approximation of $f(x) = x$ on $[-\pi, \pi]$ by $\{\sin x, \sin 2x\}$ is $2 \sin x - \sin 2x$, its first two Fourier terms. Adding further terms only decreases the $L^2$ error, which tends to 0 because the trigonometric system is complete. ◁

**Example 13.41 (A Fourier coefficient).** For $f(x) = x$ on $[-\pi, \pi]$ the coefficient against $\sin nx$ is

$$\frac{1}{\pi} \int_{-\pi}^{\pi} x \sin nx \, dx = \frac{2(-1)^{n+1}}{n},$$

evaluated by parts; summing recovers $x = 2 \sum_{n \geq 1} \frac{(-1)^{n+1}}{n} \sin nx$. ◁

**Example 13.42 (Bessel in action).** For any $v$ and orthonormal $e_1, \dots, e_N$ the partial sum $\sum_{n=1}^{N} |\langle e_n, v\rangle|^2$ never exceeds $\|v\|^2$. Equality in the limit $N \to \infty$ is exactly completeness of the basis—no "energy" left unaccounted for. ◁

**Example 13.43 (Projecting onto constants).** The orthogonal projection of $f \in L^2[0,1]$ onto the constants is its average,

$$Pf = \frac{\langle 1, f\rangle}{\|1\|^2} = \int_0^1 f,$$

the best constant approximation. For $f(x) = x$ this is $\frac{1}{2}$, with residual $x - \frac{1}{2}$ orthogonal to the constants. ◁

**Example 13.44 (A second Parseval sum).** The square wave $\operatorname{sgn}(x)$ on $[-\pi, \pi]$ expands in odd sine harmonics, $\operatorname{sgn}(x) = \frac{4}{\pi}\sum_{k\geq 0}\frac{\sin(2k+1)x}{2k+1}$, and Parseval then gives $\sum_{k\geq 0}\frac{1}{(2k+1)^2} = \frac{\pi^2}{8}$. Different functions unlock different classical sums. ◁

**Example 13.45 (Energy of a sawtooth).** For $f(x) = x$ on $[-\pi, \pi]$, direct integration gives $\|f\|^2 = \int_{-\pi}^{\pi} x^2\,dx = \frac{2\pi^3}{3}$. Parseval with the coefficients $\frac{2(-1)^{n+1}}{n}$ and $\|\sin nx\|^2 = \pi$ gives $\|f\|^2 = \pi\sum_{n\geq 1}\frac{4}{n^2}$; equating the two recovers $\sum 1/n^2 = \pi^2/6$ once more. ◁

**Example 13.46 (Least-squares fit in $L^2$).** The best linear approximation to $f(x) = x^2$ on $[-1, 1]$—its projection onto $\operatorname{span}\{1, x\}$—uses $\langle 1, x^2\rangle = \frac{2}{3}$, $\langle 1, 1\rangle = 2$, $\langle x, x^2\rangle = 0$:

$$Pf = \frac{\langle 1, x^2\rangle}{\langle 1, 1\rangle}\,1 = \frac{1}{3}.$$

The residual $x^2 - \frac{1}{3}$ is the Legendre polynomial $p_2$ of Example 13.37, orthogonal to every linear function. ◁

**Example 13.47 (Fourier series of $|x|$).** The even function $|x|$ on $[-\pi, \pi]$ expands in cosines,

$$|x| = \frac{\pi}{2} - \frac{4}{\pi}\sum_{k\geq 0}\frac{\cos(2k+1)x}{(2k+1)^2},$$

and Parseval reproduces $\sum_{k\geq 0}(2k+1)^{-4} = \pi^4/96$. Even functions use the cosine basis, odd functions the sine basis. ◁

**Example 13.48 (Building a Fourier series).** For $f(x) = x^2$ on $[-\pi, \pi]$ the cosine coefficients are

$$a_0 = \frac{1}{2\pi}\int_{-\pi}^{\pi} x^2\,dx = \frac{\pi^2}{3}, \qquad a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} x^2\cos nx\,dx = \frac{4(-1)^n}{n^2},$$

so $x^2 = \frac{\pi^2}{3} + 4\sum_{n\geq 1}\frac{(-1)^n}{n^2}\cos nx$. Setting $x = \pi$ recovers $\sum 1/n^2 = \pi^2/6$, while $x = 0$ gives $\sum(-1)^{n+1}/n^2 = \pi^2/12$. A single expansion encodes several classical sums. ◁

**Remark 13.49 (Higher-dimensional bases).** On the sphere the *spherical harmonics* $Y_{\ell m}$ are a complete orthonormal basis of $L^2(S^2)$; on $\mathbb{R}^3$ one multiplies them by radial functions. Expanding a wavefunction this way is separation of variables, and the labels $\ell, m$ are the angular-momentum quantum numbers—a complete set of commuting observables (Lecture 10).

## 5 The Riesz theorem in Hilbert space

A linear functional $\varphi\colon \mathcal{H} \to \mathbb{C}$ is *bounded* (equivalently continuous) if $|\varphi(v)| \leq C\|v\|$ for some constant $C$.

> **Theorem 13.50 (Riesz–Fréchet · cited).** Every bounded linear functional on a Hilbert space is $\varphi = \langle u, \cdot\rangle$ for a unique $u \in \mathcal{H}$, with $\|\varphi\| = \|u\|$.

So the dual of a Hilbert space is again the space (the bra $\langle u|$ is the functional $\langle u, \cdot\rangle$), exactly as in Lecture 8—but only for the *bounded* functionals. In infinite dimensions unbounded functionals exist, and the restriction to continuous ones is the price of the infinite-dimensional dual. This continuity caveat returns with full force for operators in Lecture 14.

**Example 13.51 (Bounded and unbounded functionals).** On $L^2[0, 1]$ the functional $f \mapsto \int_0^1 f$ is bounded (by Cauchy–Schwarz $|\int f| \leq \|f\|_2$), so Riesz gives it as $\langle u, \cdot\rangle$ with $u \equiv 1$. By contrast point evaluation $f \mapsto f(0)$ is *not* bounded on $L^2$—functions close in norm can differ wildly at a point—so it is no inner product against any vector. ◁

**Example 13.52 (A bounded operator).** On $L^2[0,1]$, multiplication by a bounded function, $(Mf)(x) = m(x)f(x)$, is a bounded operator with $\|M\| = \sup|m|$, and its adjoint is multiplication by $\overline{m}$. Position $\hat{x}$ is the unbounded cousin $m(x) = x$ on all of $\mathbb{R}$, which is *not* bounded—the theme of Lecture 14. ◁

**Example 13.53 (Adjoint of the shift).** The right shift $S(x_1, x_2, \dots) = (0, x_1, x_2, \dots)$ on $\ell^2$ is bounded with $\|S\| = 1$, and its adjoint is the left shift

$$S^\dagger(x_1, x_2, \dots) = (x_2, x_3, \dots),$$

since $\langle Sx, y\rangle = \langle x, S^\dagger y\rangle$. Unlike in finite dimensions, $S$ is an isometry that is not surjective—a purely infinite-dimensional phenomenon. ◁

**Example 13.54 (Riesz in $\ell^2$).** The functional $\varphi(x) = \sum_n \frac{x_n}{n}$ on $\ell^2$ is bounded, and Riesz identifies it with $u = (1, \frac{1}{2}, \frac{1}{3}, \dots) \in \ell^2$ (which lies in $\ell^2$ since $\sum 1/n^2 < \infty$), because $\varphi(x) = \langle u, x\rangle = \sum_n \overline{u_n} x_n$. The functional and its representing vector carry the same data. ◁

**Remark 13.55 (Bras, kets, and rigged spaces).** Kets are vectors in $\mathcal{H}$; bras are the bounded functionals, elements of $\mathcal{H}^* \cong \mathcal{H}$. Idealized states like the position eigenket $|x\rangle$ are *not* in $\mathcal{H}$—a plane wave is not square-integrable—and making them rigorous requires the "rigged Hilbert space" that frames the continuous spectrum of Lecture 14.

**Remark 13.56 (Toward operators).** A linear map $T : \mathcal{H} \to \mathcal{H}$ is *bounded* if $\|Tv\| \le C\|v\|$, and then has an adjoint $T^\dagger$ defined just as in Lecture 9. But the operators of physics—position, momentum, energy—are typically *unbounded*, defined only on a dense domain. Handling them with care is the work of the final lecture.

## Summary of main concepts

A metric measures distance; a norm is a metric compatible with the linear structure, $d(u,v) = \|u - v\|$. A space is complete if every Cauchy sequence converges; complete normed spaces are Banach spaces, and complete inner product spaces are Hilbert spaces. Finite-dimensional spaces are automatically complete, so completeness is an infinite-dimensional concern. The model Hilbert spaces are $\ell^2$ (square-summable sequences) and $L^2$ (square-integrable functions), the state spaces of quantum mechanics. A complete orthonormal basis gives the Fourier expansion $v = \sum_n \langle e_n, v\rangle e_n$ with Parseval's identity $\|v\|^2 = \sum_n |\langle e_n, v\rangle|^2$; Bessel's inequality holds for any orthonormal set. Finally, Riesz–Fréchet identifies the *bounded* dual of a Hilbert space with the space itself—bras are dual vectors, provided they are continuous.

- **Metric and norm**—distance; a norm is a metric compatible with the linear structure, $d(u,v) = \|u - v\|$.
- **Completeness**—every Cauchy sequence converges; Banach (normed) and Hilbert (inner product); automatic in finite dimensions.
- **$\ell^2$ and $L^2$**—square-summable sequences and square-integrable functions—the state spaces of quantum mechanics.
- **Hilbert basis**—a complete orthonormal set; Fourier expansion $v = \sum_n \langle e_n, v\rangle e_n$ with Parseval; Bessel for any orthonormal set.
- **Riesz–Fréchet**—the *bounded* dual of a Hilbert space is the space itself—bras are continuous dual vectors.
- **Why infinite dimensions differ**—the closed unit ball is no longer compact.

*Exercises for this lecture are in the companion sheet `exercises-13.pdf`.*